

hash

人员

周子一、杨瑾硕、秦显森、于子珈、牛同泽 到课, 李子瀚 线上

上周作业检查

上周作业链接: <https://cppoj.kids123code.com/contest/4675>

王向东老师周日十点半C++trie树

🔄 刷新

#	用户名	姓名	编程分	时间	A	B	C	D
1	zhouziyi	周子一	400	2244	100	100	100	100
2	lizihan	李子瀚	400	3036	100	100	100	100
3	niutongze	牛同泽	300	1207	100	100	100	
4	yangjinshuo	杨瑾硕	200	1049	100	100		
5	yuzijia1	于子珈	200	1697	100	100		
6	xieyakai	谢亚楷	200	1728	100	100		
7	qinxiansen	秦显森	200	1737	100	100		
8	slyunxin	司云心	200	2058	100	100		
9	chenluoran	陈洛冉	100	1571	100	0		

本周作业

<https://cppoj.kids123code.com/contest/4812> (课上讲了 A ~ B 题, 课后作业是 C 题必做)

课堂表现

今天给同学们讲了 hash 这个算法, 今天的 B 题比较难一些, 课上老师带着大家写了一遍, 课下同学们最好再独立写一遍。

课堂内容

P10468 兔子与兔子

```
// 单 hash
#include <bits/stdc++.h>

using namespace std;

typedef unsigned long long ULL;
const int maxn = 1e6 + 5;
const int P = 131;
char s[maxn];
ULL p[maxn], h[maxn];

ULL get_hash(int l, int r) { return h[r] - h[l-1]*p[r-l+1]; }
```

```

int main()
{
    cin >> (s+1);
    int n = strlen(s+1);

    p[0] = h[0] = 1;
    for (int i = 1; i <= n; ++i) {
        p[i] = p[i-1]*P, h[i] = h[i-1]*P + s[i];
    }

    int m; cin >> m;
    while (m -- ) {
        int l1, r1, l2, r2; cin >> l1 >> r1 >> l2 >> r2;
        if (get_hash(l1,r1) == get_hash(l2,r2)) cout << "Yes" << endl;
        else cout << "No" << endl;
    }

    return 0;
}

```

```

// 双 hash
#include <bits/stdc++.h>
#define x first
#define y second

using namespace std;

typedef long long LL;
typedef pair<int, int> PII;
const int maxn = 1e6 + 5;
const int mod1 = 1e9+7, mod2 = 1e9+9;
const int P = 131;
char s[maxn];
PII p[maxn], h[maxn];

PII get_hash(int l, int r) {
    int h1 = (h[r].x - (LL)h[l-1].x*p[r-l+1].x%mod1 + mod1) % mod1;
    int h2 = (h[r].y - (LL)h[l-1].y*p[r-l+1].y%mod2 + mod2) % mod2;
    return {h1, h2};
}

int main()
{
    cin >> (s+1);
    int n = strlen(s+1);
    p[0] = h[0] = {1, 1};
    for (int i = 1; i <= n; i++) {
        h[i].x = ((LL)h[i-1].x*P + s[i]) % mod1;
        h[i].y = ((LL)h[i-1].y*P + s[i]) % mod2;
        p[i].x = (LL)p[i-1].x * P % mod1;
    }
}

```

```

        p[i].y = (LL)p[i-1].y * P % mod2;
    }

    int m; cin >> m;
    while (m -- ) {
        int l1, r1, l2, r2; cin >> l1 >> r1 >> l2 >> r2;
        if (get_hash(l1,r1) == get_hash(l2,r2)) cout << "Yes" << endl;
        else cout << "No" << endl;
    }
    return 0;
}

```

[CSP-S 2023] 消消乐

csp-s 真题: 栈 + hash 维护

栈里面维护单个字符, 以及整个栈的 hash 值

```

#include <bits/stdc++.h>

using namespace std;

typedef long long LL;
typedef unsigned long long ULL;
const int maxn = 2e6 + 5;
const int P = 131;
char s[maxn];
struct node {
    char c;
    ULL val;
};

int main()
{
    int n; cin >> n >> (s+1);
    map<ULL, int> mp;
    ULL hVal = 1; ++mp[hVal];
    stack<node> stk; stk.push({' ', 1});
    for (int i = 1; i <= n; ++i) {
        if (!stk.empty() && s[i]==stk.top().c) stk.pop();
        else stk.push({s[i], stk.top().val*P+s[i]});

        ++mp[stk.top().val];
    }

    LL res = 0;
    for (auto it : mp) res += (LL)it.second*(it.second-1) / 2;
    cout << res << endl;
    return 0;
}

```

匹配统计

hash + 二分 维护

枚举点 i 作为起点, 针对 i 作为起点时, 二分长度

```
#include <bits/stdc++.h>

using namespace std;

typedef unsigned long long ULL;
const int maxn = 2e5 + 5;
const int P = 131;
char a[maxn], b[maxn];
ULL p[maxn], ha[maxn], hb[maxn];
int f[maxn];

ULL get_hash(int l, int r, ULL h[]) { return h[r] - h[l-1]*p[r-l+1]; }

int calc(int id, int n, int m) {
    int l = 1, r = min(n-id+1, m);
    while (l <= r) {
        int mid = (l + r) / 2;
        if (get_hash(id, id+mid-1, ha) == get_hash(1, mid, hb)) l = mid+1;
        else r = mid-1;
    }
    return r;
}

int main()
{
    int n, m, T; cin >> n >> m >> T;
    cin >> (a+1) >> (b+1);
    p[0] = ha[0] = hb[0] = 1;
    for (int i = 1; i < maxn; ++i) p[i] = p[i-1]*P;
    for (int i = 1; i <= n; ++i) ha[i] = ha[i-1]*P + a[i];
    for (int i = 1; i <= m; ++i) hb[i] = hb[i-1]*P + b[i];

    for (int i = 1; i <= n; ++i) {
        int len = calc(i, n, m);
        ++f[len];
    }

    while (T -- ) {
        int x; cin >> x; cout << f[x] << endl;
    }
    return 0;
}
```